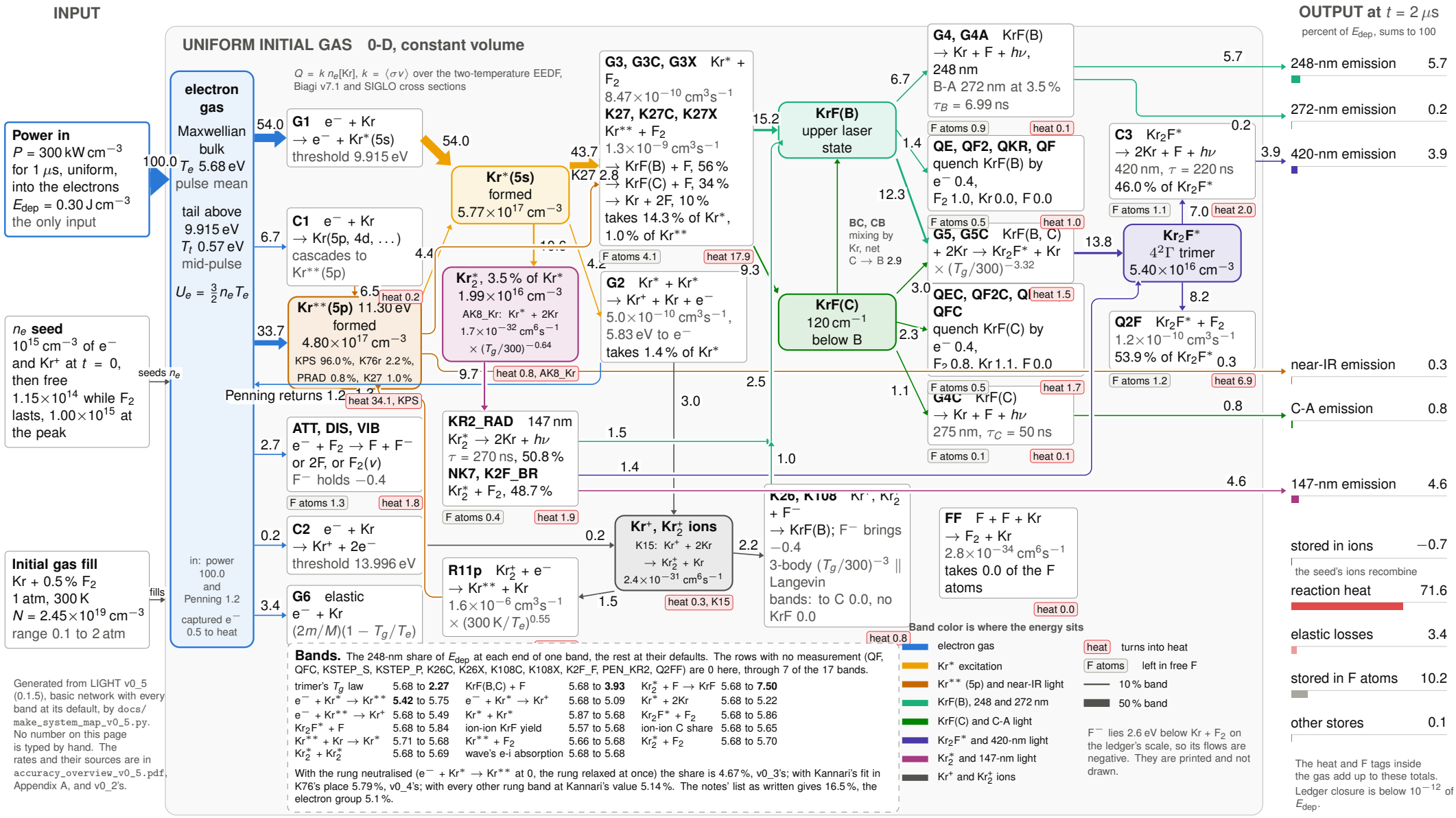
LIGHT system map, v0_5: energy flow through the Kr/F_2 kinetics with the 5p rung

Reference point: 1 atm of Kr with 0.5 % F_2 at 300 K, P = 300 kW cm^{-3} for 1 μs so that E_{dep} = 0.30 J cm^{-3}, n_e seeded at 10^{15} cm^{-3} and then free, every band at its default, run to 2 μs on v0_2's electron closure. Band width is proportional to the energy flow. Numbers are percent of E_{dep}, rounded to 0.1.



The notes' outputs as flows, which overlap: to Kr* 60.7 %, to Kr+ 3.2 %, to KrF(B) and KrF(C) 23.0 %, heat 71.6 %, elastic losses 3.4 %, emission 15.4 %. **The 5p rung.** The 5s ↔ 5p electron cycle (K76, K76r; v0_5: Jung et al.'s measured cross sections, 1.14×10^{-14} cm^2 at 6 eV, 0.92 times Kannari's fit at T_e = 5.68 eV) circulates 0.8 times the Kr* formed, so it is drawn as its two net bands: of the Kr* formed, 80.9 % is pumped up by the electrons (net of the return), the rung relaxes 96.0 % of it to Kr* on Kr (KPS, 1.7×10^{-10} cm^{3}s^{-1}; PRAD, τ = 29 ns, radiates 0.8 %), and the gap of 1.385 eV per cycle takes 33.7 % of the input as heat. Of the KrF formed, 8.63×10^{16} cm^{-3}, 3.6 % came from ion-ion recombination, 5.6 % from Kr_2^* + F_2 and 4.8 % from Kr** + F_2; 24.7 % radiates at 248 nm, 56.9 % forms the trimer. At the end T_g = 742 K, p = 2.48 atm; F_2 fell below 10 % of the fill at 835 ns. **Optics at 248** nm (`optics.py`, a diagnostic, no cavity), over 500 to 1000 ns: g_0 = 3.96, α = 0.64, $g_0 - \alpha$ = 3.32 %/cm (g_0 peaks at 5.04 at 538 ns), $\sigma_B = 2.6\times10^{-16}$ cm^2, α by species Kr_2F^* 0.44, Kr** 0.10, Kr* 0.03, F_2 0.02, Kr_2^* 0.02, F^- 0.02 %/cm, Kr_2F^* at Hakuta et al.'s measured 1.0×10^{-18} cm^2.